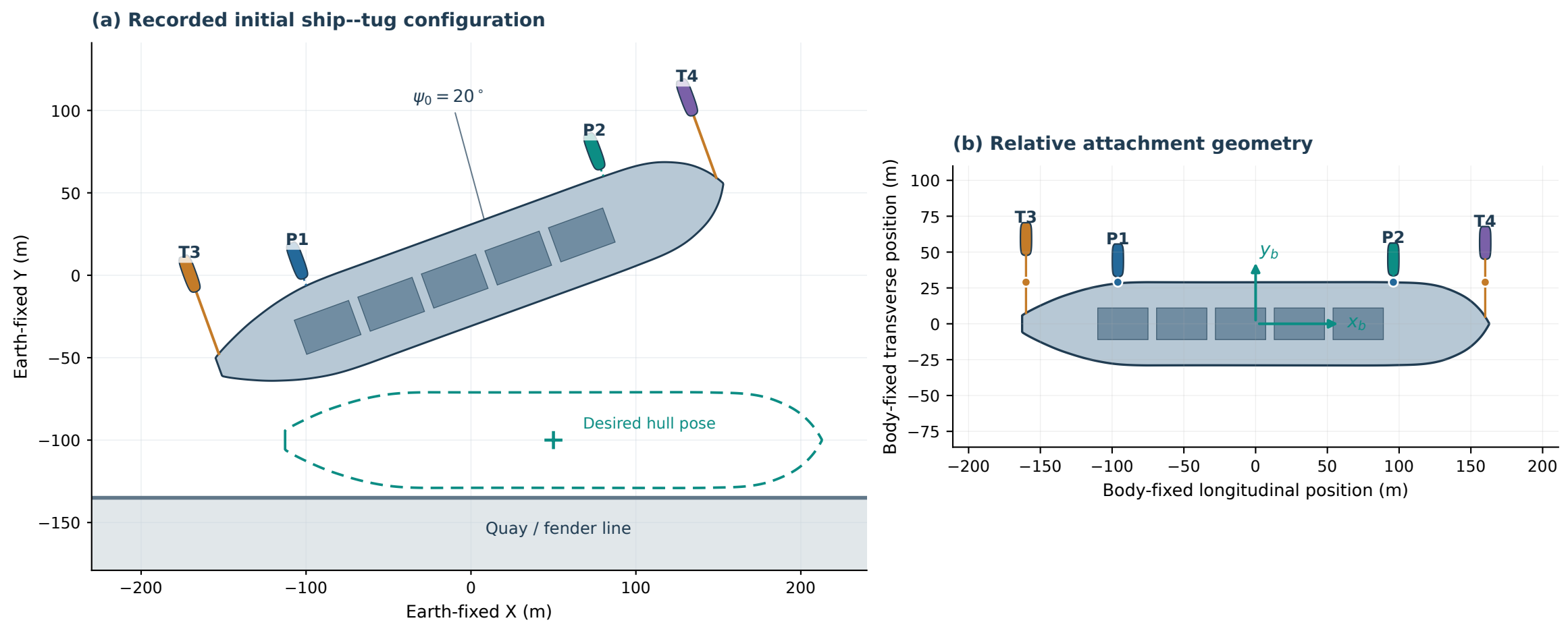

(a) Recorded initial ship--tug configuration
$\psi_0 = 20^\circ$
T4
P2
T3
P1
Desired hull pose
Quay / fender line
Earth-fixed Y (m)
Earth-fixed X (m)
100
50
0
−50
−100
−150
−200
−100
0
100
200
(b) Relative attachment geometry
T3
P1
y_b
P2
T4
x_b
Body-fixed transverse position (m)
Body-fixed longitudinal position (m)
100
75
50
25
0
−25
−50
−75
−200
−150
−100
−50
0
50
100
150
200